

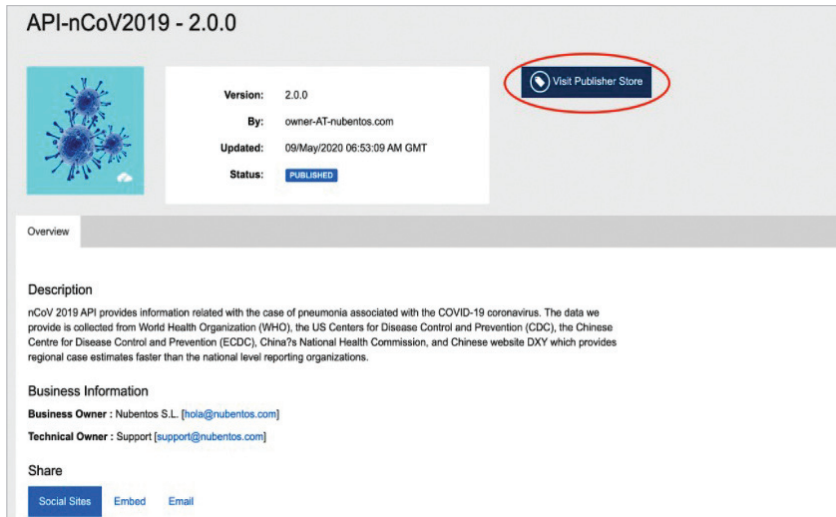


Figure 5: API-nCov19 description page

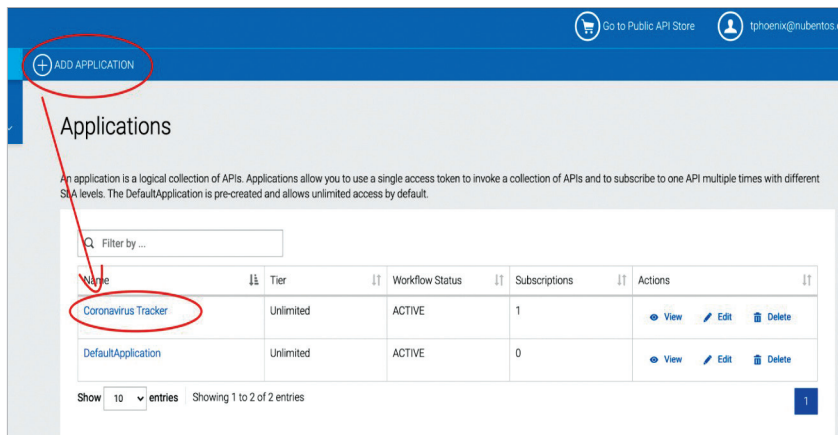


Figure 6: API-nCov19: Creating a test app on API Store

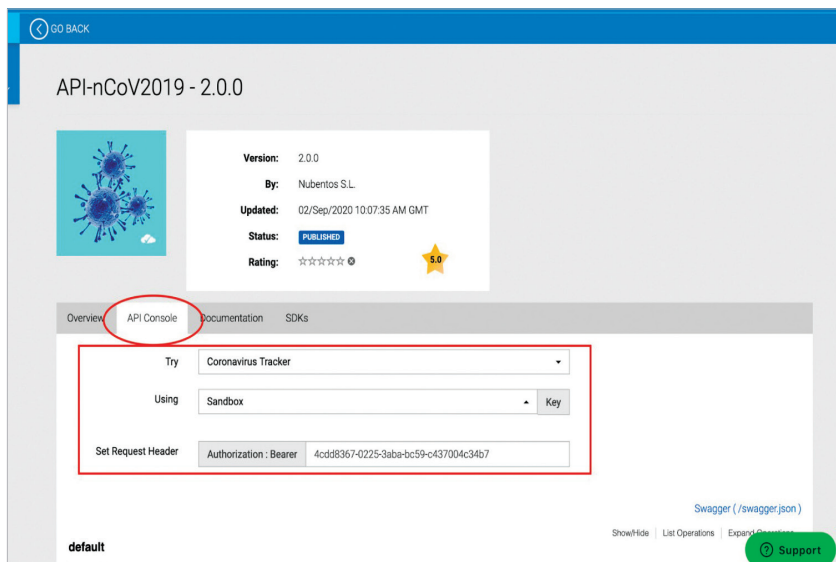


Figure 7: API-nCov19: Using API console with the test app created in the sandbox environment

developer extends his/her app. Sub-folders are created to separate functions, modals, views and widgets.

- **pubspec.yaml:** This file needs to be updated according to the app dependencies. To connect the API, this app uses the *http* library, which is updated under the *dependencies* section. Images added in *Assets* are also updated under the *flutter: assets:* section.
- Figure 10 shows the project overview and Figure 11 shows the tree structure of the Flutter Code files.

Flutter code overview

- **app.services:**
 - **api.dart:** This file contains an API class, where *host*, *port* and *basePath* are declared, along with methods to construct the uniform resource identifier (URI) address. These URI addresses are used in *api_services* methods to fetch the corresponding data.
 - **Uri tokenURI():** This builds the URI address to get an *accessToken*. Here the *query-Parameter* is `{'grant_type': 'client_credentials'}`.
 - **Uri endpointURI():** The URI address for endpoint is composed here.
 - **Uri endpointURIforQuery():** The URI address for endpoint along with *queryParameter{country}* is composed here.
 - **api_services.dart:** This file contains the *APIService* class, which contains methods for the following.
 - **Future<String> getAccessToken():** Fetches the access token that is necessary for further data queries.
 - **Future<int> getEndPointData():** This method gets the endpoint data using *accessToken* and endpoint as a parameter.